

1 • Flow conditions

$M_e, Re, T_w/T_e$, gas model

2 • Mean flow

self-similar BVP $\rightarrow \bar{U}, \bar{T}$ and y -derivatives

Section 2:
mean flow

3 • Disturbance ODEs

linearise + normal mode $\rightarrow$ Eqs. (C)–(E)

Section 3:
disturbance
equations

4 • Discretisation

Chebyshev collocation: $D \rightarrow D_1, D^2 \rightarrow D_2$

5 • Eigenvalue problem

$4(n+1) \times 4(n+1)$ pencil

Section 4:
numerical
methods

temporal: $\alpha \in \mathbb{R}$, solve c

spatial: $\omega \in \mathbb{R}$, solve α

$$A(\alpha) \phi = c B(\alpha) \phi$$

temporal growth $\omega_i = \alpha c_i$

$$(C_0 + \alpha C_1 + \alpha^2 C_2) \phi = 0$$

spatial growth $-\alpha_i$

Neutral curves

locus $c_i = 0$ in (R, α)

N-factor

$$N(R) = \int -\alpha_i dR \rightarrow e^N$$